EXPERIMENT 12

Simple Programming using REPEAT, WHILE

AIM:

To learn how to use various MySQL loop statements including while, repeat to run a block of code repeatedly based on a condition.

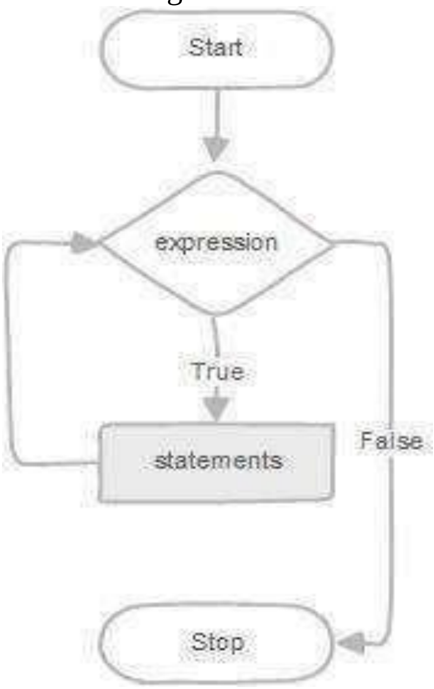
WHILE loop Syntax:

WHILE expression
DOstatements
END WHILE;

Procedure:

1. The WHILE loop checks the expression at the beginning of each iteration.
2. If the expression valuates to TRUE, MySQL will execute statements between WHILE and END WHILE until the expression evaluates to FALSE.
3. The WHILE loop is called pretest loop because it checks the expression before the statements execute.

The following flowchart illustrates the WHILE loop statement:



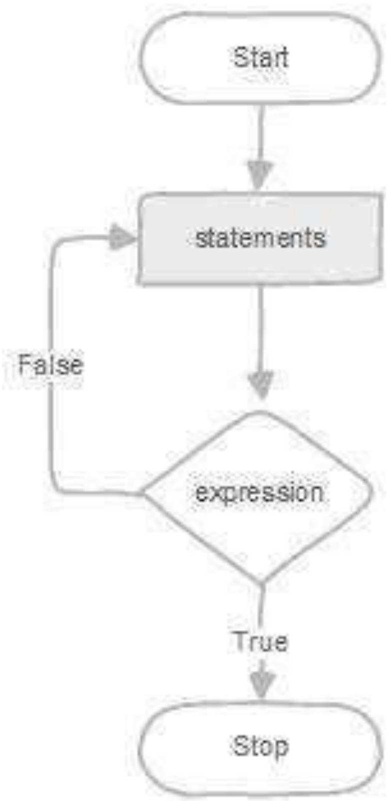
REPEAT loop Syntax:

REPEATstatements
UNTIL expression
END REPEAT;

Procedure:

1. First, MySQL executes the statements, and then it evaluates the expression.
2. Ifthe expression evaluates to FALSE, MySQL executes the statements repeatedly untilthe expression evaluates to TRUE.
3. Because the REPEAT loop statement checks the expression after the execution of statements, the REPEAT loop statement is also known as the post-test loop.

The following flowchart illustrates the REPEAT loop statement:



Questions

1. Write a MySQL program using a **WHILE loop** to display the numbers from 1 to 5.

2. Write a MySQL program using a **REPEAT loop** to display the numbers from 1 to 5.

3. Write a MySQL program using a **WHILE loop** to calculate the sum of numbers from 1 to 5.

4. Write a MySQL program using a **REPEAT loop** to calculate the sum of numbers from 1 to

OUTPUT:

1)

```
DELIMITER //

CREATE PROCEDURE DisplayNumbers()
BEGIN
    DECLARE n INT DEFAULT 1;
    DECLARE result VARCHAR(100) DEFAULT '';

    WHILE n <= 5 DO
        SET result = CONCAT(result, n, ' ');
        SET n = n + 1;
    END WHILE;

    SELECT result AS Numbers;
END //

DELIMITER ;

CALL DisplayNumbers();
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Numbers
1 2 3 4 5

2)

```sql
DROP PROCEDURE IF EXISTS DisplayNumbers;
DELIMITER //
CREATE PROCEDURE DisplayNumbers()
BEGIN
    DECLARE n INT DEFAULT 1;
    DECLARE result VARCHAR(100) DEFAULT '';

    REPEAT
        SET result = CONCAT(result, n, ' ');
        SET n = n + 1;
    UNTIL n > 5
    END REPEAT;

    SELECT result AS Numbers;
END //
DELIMITER ;
CALL DisplayNumbers();
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Numbers
1 2 3 4 5

3)

```sql
CREATE PROCEDURE CalculateSum()
BEGIN
    DECLARE n INT DEFAULT 1;
    DECLARE total INT DEFAULT 0;

    WHILE n <= 5 DO
        SET total = total + n;
        SET n = n + 1;
    END WHILE;

    SELECT total AS Sum_of_Numbers;
END //

DELIMITER ;

CALL CalculateSum();
```

Result Grid | Filter Rows: | Export: | W

Sum_of_Numbers
15

y

4)

```
CREATE PROCEDURE CalculateSumRepeat()
BEGIN
    DECLARE n INT DEFAULT 1;
    DECLARE total INT DEFAULT 0;

    REPEAT
        SET total = total + n;
        SET n = n + 1;
    UNTIL n > 5
    END REPEAT;

    SELECT total AS Sum_of_Numbers;
END //

DELIMITER ;

CALL CalculateSumRepeat();
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conte

Sum_of_Numbers
15

RESULT: Thus the Simple programming exercise using (REPEAT, WHILE)executed successfully.